SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Department of Artificial Intelligence and Machine Learning
Big Data Analytics Poster AY 2026-27

Real-Time Credit Card Fraud Detection using Apache Spark Streaming

Abstract

- Credit-card networks produce high-velocity, highly imbalanced transaction streams that batch analytics cannot score in time to block fraud.
- This project builds an Apache Spark Structured Streaming and MLlib pipeline on the ULB Credit Card Fraud dataset (284,807 transactions; only 0.172% fraud).
- A class-weighted Random Forest flags suspicious payments in 2-second micro-batches with AUC-ROC 0.968 and F1-score 0.84, enabling near real-time bank alerts.

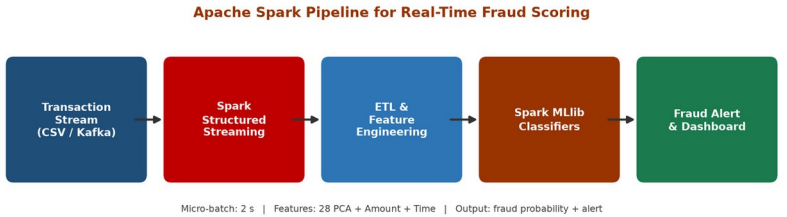
Project Objectives/Aim

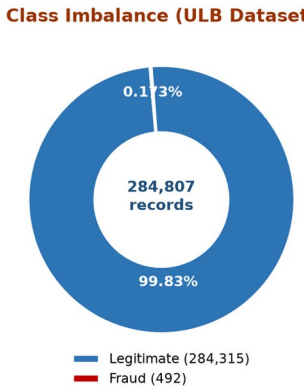
- Ingest and clean large-scale, imbalanced card-transaction data using Apache Spark.
- Engineer amount, time and PCA-based features that capture rare fraud patterns.
- Train and compare Logistic Regression, Random Forest and GBT models in Spark MLlib.
- Deploy a Structured Streaming scorer with micro-batch latency under two seconds.

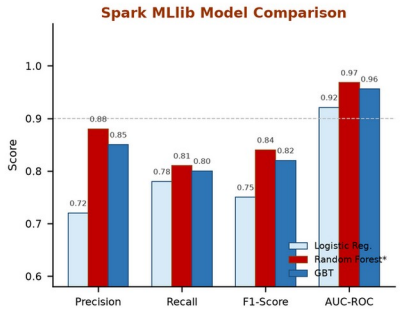
Project Working/Methodology

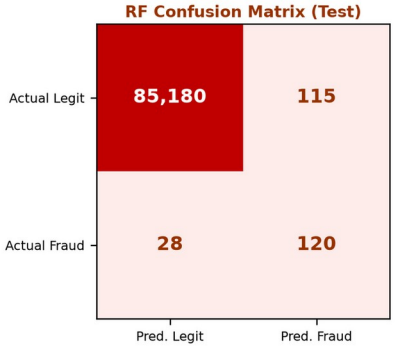
- Dataset: ULB/Kaggle Credit Card Fraud (Sept 2013; 31 features — Time, Amount, V1–V28 PCA components).
- ETL: Spark DataFrame load, null handling, StandardScaler, 70:30 stratified split and class-weighting for the 0.172% fraud rate.
- Models: Spark MLlib Logistic Regression, Random Forest (100 trees) and Gradient-Boosted Trees, evaluated with Precision, Recall, F1 and AUC-ROC.
- Streaming: Structured Streaming (2 s micro-batches) applies the fitted pipeline and emits fraud probability plus an alert to a monitoring dashboard.
- Stack: Apache Spark 3.5, PySpark, Spark MLlib and Python 3.11 on a local cluster-style workflow.

Screen Shot photos/ Results/Outcome









References: (APA Style)

Dal Pozzolo, A., et al. (2015). IEEE SSCI. Carcillo, F., et al. (2018). Information Fusion, 41, 182–194.
Apache Software Foundation. (2024). Spark MLlib guide. https://spark.apache.org/docs/latest/ml-guide.html

Student Name (PRN): Sukhada Dhingra (24070126505)

Programme: B.Tech — Artificial Intelligence and Machine Learning | AY 2026–27

Subject Teacher: Dr. Archana Choudary